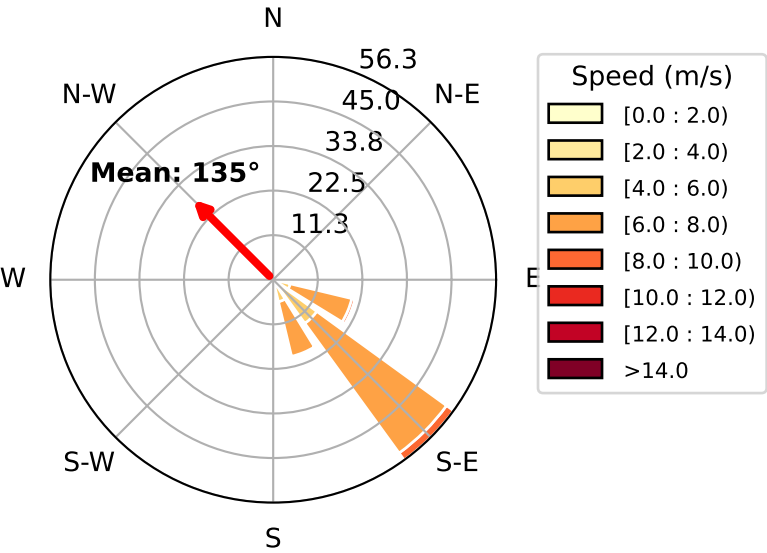
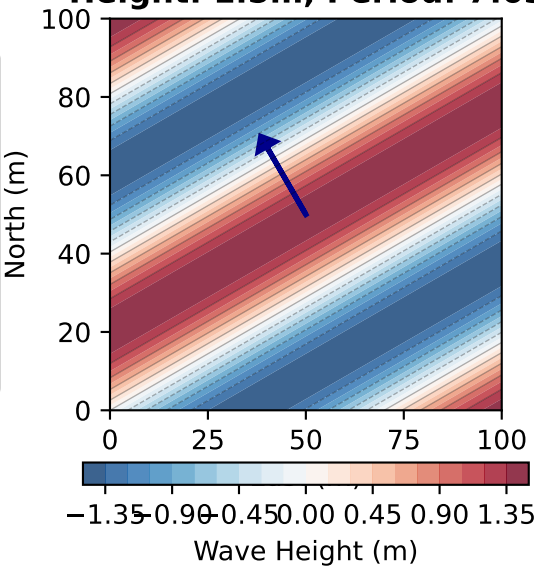


# Environmental Conditions and Virtual Actuator Analysis

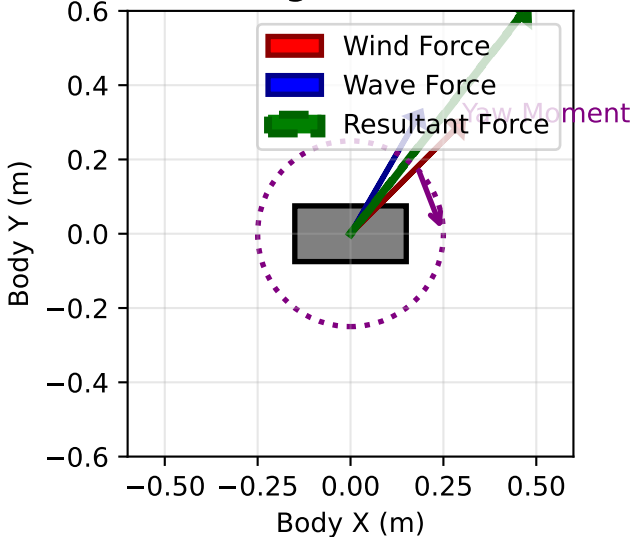
**Wind Conditions  
(Sea State 4)**



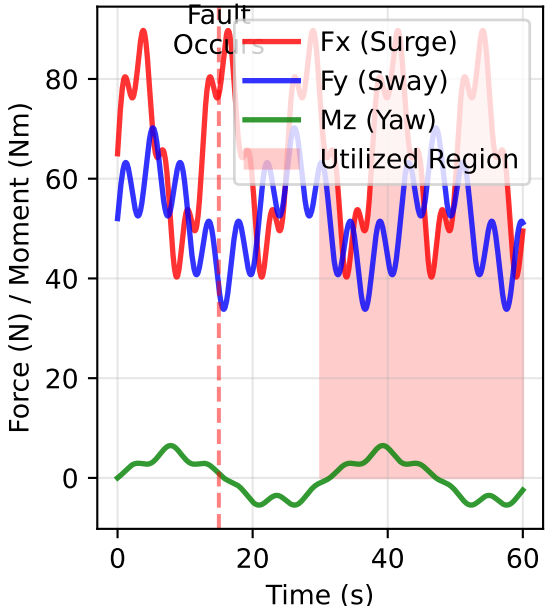
**Wave Field  
Height: 1.5m, Period: 7.0s**



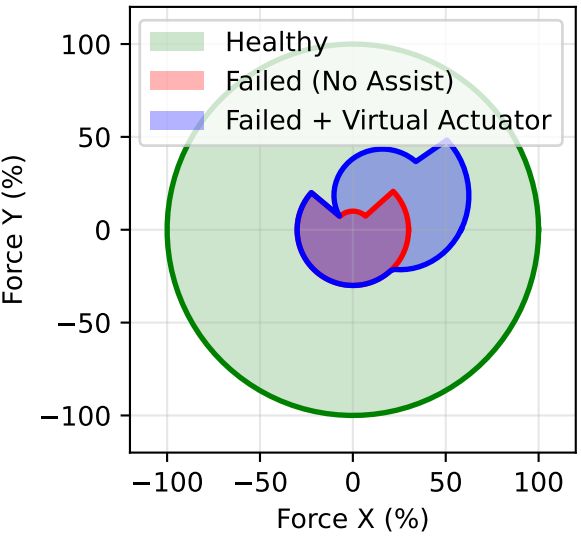
**Environmental Forces on USV  
(15° Force Misalignment → Yaw Moment)**



**Environmental Forces Over Time**



**Control Authority Envelope  
(95% Thruster Failure)**



**Virtual Actuator Strategy**

